

Lab 3 — Build a Live Data Robot · Tonight in 20 Steps

OPIM 5512 · Module 3 — Operational ML

The one-sentence version: tonight you build two tiny APIs and hand them to a robot that collects live energy + weather data every 5 minutes — forever, for free, while you sleep. Partner A builds the **energy** API (ISO-NE), Partner B builds the **weather** API (National Weather Service), and you merge them through the same branch → PR → review → merge loop as Labs 1 and 2.

Before we start

- [] **GitHub account + GitHub Desktop** signed in · **VS Code** opens with the **Python** and **Jupyter** extensions
- [] Laptop Python fighting you? Skip the fight: **Codespaces** (your repo → **Code** → **Codespaces** → **Create**) is VS Code in the browser with everything pre-installed. Same steps.
- [] Roles picked: **Partner A = energy** (`src/isone.py`), **Partner B = weather** (`src/nws.py`)
- [] Your job is **3 small TODOs** (A) or **2 small TODOs** (B). The robot and the tests are already written.

⚠ The 3 things that trip everyone up (read this first)

1. **Partner B must ACCEPT the email invite before anything works.** Check inbox and spam the moment Partner A adds you.
2. **Make your dev- branch BEFORE you touch any file.** Already edited on `main` by accident? **New branch** → "Bring my changes to the new branch" → **yes**. Nothing lost.
3. **Your repo must be PUBLIC.** Public repos get unlimited free robot minutes. A private repo would burn through its free minutes in about a week.

Flying solo? You are both A and B: set approvals to **0**, make **two branches** (`dev-energy` , `dev-weather`), build energy first, then weather.

The 20 steps, in order

Set up (steps 1–6)

1. **Pair up:** Partner A (energy) / Partner B (weather).
2. **Partner A:** open the [template repo](#) → **Use this template** → **Create a new repository** → owner = *you*, name `opim5512-lab3-<netidA>-<netidB>`, **Public** → Create.
3. **Partner A: Settings** → **Collaborators** → **Add people** → Partner B. → **STOP: Partner B accepts the invite now.**
4. **Partner A: Settings** → **Rules** → ruleset on `main`: **require a pull request + 1 approval**. (*Solo? approvals = 0.*)
5. **Both:** GitHub Desktop → **File** → **Clone repository** → pick the repo → **Clone**. Once.

6. **Each:** **Current branch** → **New branch** → `dev-energy` (A) / `dev-weather` (B) → **Publish branch** → **Fetch/Pull**.

Build your API (steps 7–12)

7. **Each:** GitHub Desktop → **Repository** → **Open in Visual Studio Code**. Check the top bar still says your `dev-` branch.
8. **Each:** open **your** file — `src/isone.py` (A) or `src/nws.py` (B). Fill in the **TODOs** (the comments tell you exactly what goes there), then **delete the** `raise NotImplementedError` **line**. (*Stamford, Partner B: also change* `STATION = "KBDL"` **to** `"KBDR"`.)
9. **Each:** VS Code terminal → `pip install -r requirements.txt` → `python src/isone.py` (A) or `python src/nws.py` (B). **You should see real rows from right now**.
10. **Each:** GitHub Desktop → top bar = your `dev-` branch → Summary `build the energy API` / `build the weather API` → **Commit** → **Push**.
11. **Each:** github.com → **Compare & pull request** → **Create** → **Reviewers** → your partner. **Watch the check run:** ✓ green = your tests passed (your partner's show as *skipped* until they merge).
12. **Each:** open your **partner's** PR → **Files changed** → read their function + look for the green ✓ → **Approve**.

Merge & wake the robot (steps 13–16)

13. **Merge** both PRs → **Delete branch** on each.
14. **Actions** tab → **collect** → **Run workflow** → Run. (*It also runs by itself every ~5 minutes — this just skips the wait.*)
15. When the run turns ✓, go to **Code** → switch the branch menu from `main` to `data` → open `isone_5min.csv` and `weather_obs.csv`. **That's your robot's work**.
16. **Both:** GitHub Desktop → `main` → **Fetch** → **Pull**. (*Your laptop gets the merged code; the data stays on the* `data` *branch.*)

Watch it live (steps 17–20)

17. **VS Code** (on `main` now) → open `notebooks/Lab3_Watch_Your_Live_Data.ipynb` → **Select Kernel** (top right → your Python) → set `OWNER` and `REPO` → **Run All**.
18. **Look at the solar gap plot:** when is rooftop solar hiding the most demand from the grid?
19. **Wait an hour, run the notebook again — more rows.** Tomorrow — a lot more. The robot doesn't need you.
20. **Submit:** screenshot **Insights** → **Network** (the two-way loop) **and** the `data` branch's commit list (the robot's work). Read-out: one sentence each — what does your API hide, and what surprised you in the live data?

Good Advice

- **python not found (Windows)?** Try `py` instead: `py src/isone.py`, `py -m pip install -r requirements.txt`. Still stuck after 5 minutes? **Switch to Codespaces** and keep going — don't burn the lab on setup.
- **Notebook asks for a kernel or says** `ipykernel is missing?` Run `pip install -r requirements.txt` in the VS Code terminal, then **Select Kernel** again.
- **Your PR shows a red X?** Click **Details** → scroll to the red `FAILED` line. It tells you which check failed and why (e.g., `"temp_f=16.0 but the API says 15 C = 59.0 F"` means you forgot `c_to_f`).

- **Your partner's tests say *skipped*?** That's normal until they merge. Skipped ≠ failed.
 - **NotImplementedError ?** You filled in the TODOs but didn't delete the `raise NotImplementedError` line.
 - **Accidentally committed or dragged files onto `main` ?** New branch → "**Bring my changes to the new branch**" → yes.
 - **Robot run says "No new rows this run"?** That's correct, not broken. The feed hadn't published anything newer yet. It catches up next run.
 - **Weather rows look ~20 minutes old?** Normal. The weather service reports every 5 minutes but publishes about 20 minutes late. ISO-NE is 1–4 minutes late.
 - **Add your partner as Reviewer on every PR.** No reviewer = no approval = no loop.
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Definition of done — what's in the repo at 7:30

- [] Repo is **public**; both partners are collaborators; **branch protection** on `main`
- [] **2 merged pull requests** (one each), both with a green ✓, branches deleted
- [] The **data branch** exists with **both** `isone_5min.csv` and `weather_obs.csv`
- [] A **network graph** showing the loop going both ways
- [] (*optional bonus*) The notebook's solar-gap plot, with one sentence on what it shows

You didn't just analyze data tonight. You built the thing that collects it.